

ELAKKIYA SELVARAJAN

React Native Engineer | Frontend Developer | Mobile (Android & iOS)

Chennai, India | +91-7339120466 | selvarajanelakkiya@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [Medium](#)

PROFESSIONAL SUMMARY

Results-driven React Native Engineer with 2+ years of production experience shipping cross-platform mobile apps to the Google Play Store and Apple App Store. Built and scaled apps at an IIT Madras incubated startup across the full mobile development lifecycle — from architecture design through App Store release. Delivered measurable impact: 30% fewer QA defects, 25% faster feature delivery, and 40% improvement in render performance. Deep expertise in TypeScript, React Native CLI, native module bridging (Swift/Kotlin), offline-first architecture, Redux Toolkit, and CI/CD pipelines. Currently expanding into full-stack web development with React.js, Next.js, and NestJS.

TECHNICAL SKILLS

Mobile: React Native CLI, Expo, TypeScript, JavaScript, React Navigation, AsyncStorage, Native Module Bridging

Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS, Responsive Web Design

State Management: Redux, Redux Toolkit, Context API

Backend & Databases: NestJS, REST APIs, Axios, SQLite, PostgreSQL

Native & Platform: Kotlin, Swift, Objective-C Bridge, WiFi Direct, TCP Sockets, Offline-First Architecture

Tools: Jest, Git, GitHub, Postman, Android Studio, Xcode, VS Code, Jira

Practices: Agile/Scrum, CI/CD, Performance Optimization, Cross-Platform Development, Code Review

PROFESSIONAL EXPERIENCE

React Native Engineer & Frontend Developer

Oct 2023 - Present

Plenome Technologies Pvt. Ltd. (formerly ICSR) · IIT Madras Incubated

Startup · Chennai, India

Aayush – Health & Wellness App (Live on Google Play Store & Apple App Store)

- Architected and shipped 10+ production screens for Android and iOS using React Native CLI and TypeScript, reducing QA-reported defects by 30% through reusable component design patterns.
- Engineered 20+ reusable TypeScript UI components, cutting new feature development time by 25% while maintaining a consistent cross-platform design system.
- Integrated 6+ REST APIs via Axios with centralized error handling, retry logic, and loading state management, improving data reliability and perceived performance.
- Implemented Apple Watch integration via native module bridging (Swift + Objective-C bridge), enabling real-time health metric sync between iPhone and Watch.
- Applied React.memo, useMemo, and useCallback across critical render paths — reduced unnecessary re-renders by 40% and improved frame rates on low-end Android devices.
- Led React Native version migration from v0.73 to v0.81, resolving 15+ breaking dependency conflicts to maintain full Android and iOS compatibility.
- Built secure custom form validation across 8+ modules using TypeScript, reducing invalid submissions and improving input reliability.
- Debugged platform-specific runtime crashes, native bridge errors, and rendering issues across Android and iOS build environments.
- Delivered features consistently within a 5-member Agile team using Git, GitHub, and Jira with bi-weekly sprints.

Offline eRupee – Offline Peer-to-Peer Digital Payment SDK

- Designed a proof-of-concept offline P2P digital payment system enabling eRupee transactions without internet connectivity using React Native and TypeScript.
- Built an Android native module in Kotlin leveraging WiFi Direct for peer device discovery and TCP socket-based data transfer, integrated via NativeModules bridge.

- Implemented SQLite offline transaction queue with automatic cloud sync on network restoration, ensuring zero data loss across sessions.
- Architected a modular, distributable React Native SDK with a clean public API for seamless integration into external apps as a local package dependency.
- Designed cryptographic transaction validation workflows to prevent duplicate and invalid offline payments.
- Configured Android Gradle multi-module build system with React Native autolinking for SDK packaging and distribution.

React Native Developer – Intern

Feb 2023 - Sep 2023

Plenome Technologies Pvt. Ltd. (formerly ICSR) · Chennai, India

- Developed responsive UI components and screens using React Native, HTML5, CSS3, and JavaScript, meeting design specifications and cross-platform compatibility requirements.
- Validated REST APIs using Postman, ensuring smooth frontend-backend integration during feature development cycles.
- Collaborated with senior developers to debug UI issues, implement design feedback, and deliver production-ready frontend features.

Web Developer – Intern

Jun 2021 - Nov 2021

Compunet Connections · Chennai, India

- Built and maintained responsive web pages using HTML5, CSS3, and JavaScript, adhering to design specifications and cross-browser compatibility standards.
- Supported REST API integration and validation using Postman in collaboration with backend developers.

PROJECTS

Finguard AI – Full Stack AI Financial Tracker – Next.js · Tailwind CSS · NestJS · PostgreSQL · Ollama ·

TypeScript Mar 2026 - Present

- Building a full-stack AI-powered personal finance platform with intelligent expense analysis and financial insights using HuggingFace AI models.
- Designing responsive dashboards with Next.js and Tailwind CSS; architecting RESTful backend services with NestJS and PostgreSQL for scalable data management.
- Implementing personalized AI recommendations for spending pattern analysis, enabling data-driven financial decisions.

EDUCATION

M.Sc. (Integrated) – Computer Science

Aug 2018 - Jul 2023

College of Engineering, Guindy (CEG), Anna University, Chennai | CGPA: 7.67 / 10.0 | 5-Year Integrated Master's Program

CERTIFICATIONS

- Applied Database System Engineering – IITM Pravartak | May – Jul 2025 | Distributed databases, kernel-level processing, query optimization
- React Native – The Practical Guide – Udemy | March 2026 | React Native, Cross-Platform Mobile Development
- AI & ML Career with Generative AI – GUVI x HCL | June 2026 | Generative AI skills, Future-proof AI & ML career strategies
- AI Fluency: Framework & Foundations – Anthropic | June 2026 | AI frameworks, Responsible AI, Effective AI usage, AI decision-making
- Jest JavaScript Testing Framework – TestAutomationU | June 2026 | Unit testing, Async/Await testing, Matchers, beforeEach/afterEach hooks, Test automation with JavaScript